Code Coverage Report - Hostel Management System

1. Introduction

1.1 Purpose

This report gives the statement (line) coverage of the Hostel Management System achieved by the automated test suite, as required by Mini-Project Deliverables Part-3.

1.2 Scope

It covers the six Django apps: accounts, students, hostels, rooms, allocations and dashboard. The STP target is 80% line coverage of models and services.

1.3 References

- Software Requirements Specification (SRS) - Hostel Management System
- Software Architecture and Design Specification (SAD) - Hostel Management System
- Software Test Plan (STP) - Hostel Management System

2. Method

Coverage was measured with coverage.py during the final test run, using the same command as the CI pipeline:

```
coverage run --branch
--source=hostels,rooms,allocations,dashboard,accounts,students manage.py test
```

Application code is the six Django apps without their test files and migrations. Migrations are generated by Django and test files exercise themselves, so this is the figure that shows how well the system is tested. The CI pipeline measures the whole source tree, including tests, and fails below 80%.

2.1 Test Environment

Item	Details
Hardware	Team laptop (16 GB RAM, Intel i5 class)
Software	Windows 11, Python 3.11.9, Django 5.2.19, SQLite 3
Tools	Django test framework and test Client, coverage.py 7.17.2, GitHub Actions

3. Summary

Measure	Statements	Covered	Missed	Coverage
Application code	694	694	0	100.0%
Whole source tree (CI figure)	2383	2378	5	99.8%
Target (STP, CI)				80%

4. Coverage by Module

Module	Statements	Covered	Missed	Coverage
accounts	132	132	0	100.0%
allocations	244	244	0	100.0%
dashboard	19	19	0	100.0%
hostels	57	57	0	100.0%
rooms	103	103	0	100.0%
students	139	139	0	100.0%

5. Coverage by File

File	Statements	Missed	Coverage	Missing lines
accounts/__init__.py	0	0	100%	-
accounts/decorators.py	25	0	100%	-
accounts/forms.py	24	0	100%	-
accounts/services.py	22	0	100%	-
accounts/urls.py	3	0	100%	-
accounts/validators.py	36	0	100%	-
accounts/views.py	22	0	100%	-
allocations/__init__.py	0	0	100%	-
allocations/forms.py	26	0	100%	-
allocations/models.py	42	0	100%	-
allocations/services.py	78	0	100%	-
allocations/urls.py	4	0	100%	-
allocations/views.py	94	0	100%	-
dashboard/__init__.py	0	0	100%	-
dashboard/urls.py	4	0	100%	-
dashboard/views.py	15	0	100%	-
hostels/__init__.py	0	0	100%	-
hostels/forms.py	13	0	100%	-
hostels/models.py	11	0	100%	-
hostels/urls.py	4	0	100%	-
hostels/views.py	29	0	100%	-
rooms/__init__.py	0	0	100%	-
rooms/api.py	10	0	100%	-
rooms/forms.py	8	0	100%	-
rooms/models.py	33	0	100%	-
rooms/urls.py	4	0	100%	-
rooms/views.py	48	0	100%	-
students/__init__.py	0	0	100%	-
students/admin.py	6	0	100%	-
students/forms.py	37	0	100%	-

File	Statements	Missed	Coverage	Missing lines
students/models.py	25	0	100%	-
students/services.py	25	0	100%	-
students/urls.py	4	0	100%	-
students/views.py	42	0	100%	-

6. Lines Not Covered

No line of application code is left uncovered. The only uncovered lines are in test files:

File	Lines	Reason
accounts/tests.py	193, 204, 228, 239	Bodies of the dummy views used to test the role decorators. The decorator rejects the request before the view runs, which is exactly what those tests check.
allocations/tests.py	290	The 'gave up' fallback in UT_ALLOC_05 for a thread that never gets the database lock. Both threads always finish, so it is not reached.

7. Conclusion

Line coverage of the application code is 100.0% and of the whole source tree 99.8%, above the 80% target in the STP.